

Task 5: Join Queries Equivalent and Recursive Queries

AIM: To implement and execute joins, equivalent queries and recursive queries in SQL

Procedure:

1. Create the table customer and Hotel
2. Insert the values into tables
3. Perform join operation
4. Perform Equivalent and recursive query
5. Display Result

```
CREATE TABLE Customer(
```

```
    SSN VARCHAR(20) PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Email VARCHAR(100) UNIQUE,  
    Phone VARCHAR(15) CHECK(LENGTH(PHONE)=10),  
    Address VARCHAR(200) DEFAULT 'Unknown'
```

```
);
```

```
CREATE TABLE HOTEL(
```

```
    Hotel ID INT PRIMARY KEY,  
    HotelName VARCHAR(100),  
    SSN VARCHAR(20),
```

```
);
```

```
INSERT INTO Customer  
VALUES
```

```
('C001', 'Alice', 'alice@gmail.com', '9876543210', 'New York'),  
( 'C002', 'Bob Smith', 'bobsmith@gmail.com', '9123456789', 'Los Angeles'),  
( 'C003', 'Charlie', 'charlie@gmail.com', '9887765543', 'Chicago');
```

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INSERT INTO HOTEL

VALUES

(101, 'Grand Palace', 'C001'),
(102, 'Ocean view', 'C002'),
(103, 'City Inn', 'C003');

SELECT * FROM Customer;

SSN	Name	Email	Phone	Address
C001	Alice	alice@gmail.com	9876543210	New York
C002	Bob Smith	Bobsmith@gmail.com	9123456789	Los Angeles
C003	Charlie	charlie@gmail.com	988765412	Chicago

SELECT * FROM HOTEL

Hotel ID	Hotel Name	SSN
101	Grand Palace	C001
102	Ocean view	C002
103	City Inn	C003

INNER JOIN

SELECT c.SSN, c.Name, h.Hotel Name

FROM Customer c

INNER JOIN Hotel h

ON c.SSN = h.SSN

SSN	Name	Hotel Name
C001	Alice	Grand Palace
C002	Bob Smith	Ocean view
C003	Charlie	City Inn

Left Join

```
SELECT c.SSN, c.Name, h.Hotel Name
FROM customer c
LEFT JOIN Hotel h
ON c.SSN = h.SSN;
```

SSN	Name	Hotel Name
C001	Alice	Grand Palace
C002	Bob Smith	Ocean View
C003	Charlie Brown	City Inn

Right Join

```
SELECT c.SSN, c.Name, h.Hotel Name
FROM customer c
RIGHT JOIN Hotel h
ON c.SSN = h.SSN
```

SSN	Name	Hotel Name
C001	Alice	Grand Palace
C002	Bob Smith	Ocean view
C003	Charlie Brown	City Inn

Full Outer Join

```
SELECT c.SSN, c.Name, h.Hotel Name
FROM customer c
FULL OUTER JOIN Hotel h
ON c.SSN = h.SSN;
```

SSN	Name	Hotel Name
C001	Alice	Grand Palace
C002	Bob Smith	Ocean view
C003	Charlie	City Inn

Equivalent queries

Using Join

```
SELECT C.SSN, C.Name
FROM Customer C
Join Customer Hotel H on C.SSN = h.SSN
WHERE h.hotel name = 'Grand Palace'
```

SSN	Name
C001	Alice

Recursive Queries

WITH RECURSIVE Subordinates AS (

```
SELECT SSN, Name, Referred By
FROM Customer
WHERE Referred By = 'C001'
```

UNION ALL

```
SELECT C.SSN, C.Name, C.Referred By
FROM Customer C
```

INNER JOIN ~~Referred By~~ Tree ~~on~~ C.Referred By = T.SSN

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SELECT * FROM Referred Tree;

SSN	Name	Referred by
C002	Bob Smith	C001
C003	Charlie Brown	C002

VELTECH	
EX No.	05
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
PIVA VOCE (5)	5
RECORD (5)	1
TOTAL (20)	15
	1619

1619
RESULT: This implementation of join queries, equivalent and recursive queries has successfully executed and verified